$$\Rightarrow 2x + k = 5$$

$$x = \frac{1}{2}(5 - k)$$

$$x = \frac{1}{2}(5 - k),\ y = k \text{ for all } k \in R$$

Thus by giving different values for k, we get different solution. Hence the system has infinite number of solutions.

Example 1.11

Show that the equations $3x - 2y = 6$, $6x - 4y = 10$ are inconsistent.

Solution:

The matrix equation corresponding to the given system is

$$\begin{pmatrix} 3 & -2 \\ 6 & -4 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} 6 \\ 10 \end{pmatrix}$$

$$AX = B$$

Matrix A	Augmented matrix $[A,B]$	Elementary Transformation
$\begin{pmatrix} 3 & -2 \\ 6 & -4 \end{pmatrix}$	$\begin{pmatrix} 3 & -2 & 6 \\ 6 & -4 & 10 \end{pmatrix}$	
$\sim \begin{pmatrix} 3 & -2 \\ 0 & 0 \end{pmatrix}$	$\sim \begin{pmatrix} 3 & -2 & 6 \\ 0 & 0 & -2 \end{pmatrix}$	$R_2 \to R_2 - 2R_1$
$\rho(A) = 1$	$\rho([A,B]) = 2$	

$$\therefore \rho([A,\ B]) = 2, \qquad \rho(A) = 1$$

$$\rho(A) \neq \rho([A,B])$$

$\therefore$ The given system is inconsistent and has no solution.

Example 1.12

Show that the equations

$2x + y + z = 5$, $x + y + z = 4$, $x - y + 2z = 1$ are consistent and hence solve them.

Solution:

The matrix equation corresponding to the given system is

$$\begin{pmatrix} 2 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & -1 & 2 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 5 \\ 4 \\ 1 \end{pmatrix}$$

$$A \quad X = B$$

Augmented matrix $[A,B]$	Elementary Transformation
$\begin{pmatrix} 2 & 1 & 1 & 5 \\ 1 & 1 & 1 & 4 \\ 1 & -1 & 2 & 1 \end{pmatrix}$	
$\sim \begin{pmatrix} 1 & 1 & 1 & 4 \\ 2 & 1 & 1 & 5 \\ 1 & -1 & 2 & 1 \end{pmatrix}$	$R_1 \leftrightarrow R_2$
$\sim \begin{pmatrix} 1 & 1 & 1 & 4 \\ 0 & -1 & -1 & -3 \\ 0 & -2 & 1 & -3 \end{pmatrix}$	$R_2 \to R_2 - 2R_1$ $R_3 \to R_3 - R_1$
$\sim \begin{pmatrix} 1 & 1 & 1 & 4 \\ 0 & -1 & -1 & -3 \\ 0 & 0 & 3 & 3 \end{pmatrix}$	$R_3 \to R_3 - 2R_2$
$\rho(A) = 3,\ \rho([A,B]) = 3$	

Obviously the last equivalent matrix is in the echelon form. It has three non-zero rows.

$\rho(A) = \rho([A,B]) = 3 =$ Number of unknowns .

The given system is consistent and has unique solution.

To find the solution, let us rewrite the above echelon form into the matrix form.

$$\begin{pmatrix} 1 & 1 & 1 \\ 0 & -1 & -1 \\ 0 & 0 & 3 \end{pmatrix}\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 4 \\ -3 \\ 3 \end{pmatrix}$$

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